



LICENSE AGREEMENT

Altura Rally Computer

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The Altura Rally Computer is designed exclusively for motorsport and competitive use. It is not approved for road use under standard vehicle regulations. The authors accept no liability for consequences arising from misuse.

PLEASE READ THIS AGREEMENT CAREFULLY BEFORE POWERING ON OR USING THE ALTURA RALLY COMPUTER. BY ACTIVATING, INSTALLING, OR OTHERWISE USING THIS DEVICE, YOU ("USER") AGREE TO BE LEGALLY BOUND BY ALL TERMS BELOW. IF YOU DO NOT AGREE, DO NOT POWER ON THE DEVICE AND RETURN IT IMMEDIATELY IN ITS ORIGINAL UNOPENED PACKAGING.

1 Purpose and Intended Use

The Altura Rally Computer ("the Device") is a high-precision GNSS-based odometer and navigation interface developed specifically for motorsport and competition environments.

Exclusive Use. The Device is intended exclusively for organized racing, rallying, and off-road competition. It is not intended as a primary navigational aid on public roads or as a safety-critical timing system in any context.

Regulatory Compliance. It is the User's sole responsibility to ensure installation and use comply with all applicable local, state, federal, and event-sanctioning-body regulations.

Road Use. The Device is not homologated for road use. If operated on public roads — including liaison sections, recce runs, or transit stages — the User must ensure: (a) full compliance with all applicable traffic laws; (b) that installation does not interfere with primary instruments or obstruct the driver's field of vision; and (c) compliance with all applicable national and regional legal requirements. Altura accepts no liability for consequences arising from use on public roads or outside sanctioned competition.

2 No Guarantee of Precision — Information Only

While the Device employs advanced GNSS signal processing and filtering techniques, fundamental limitations inherent in satellite-based positioning apply at all times.

Informational Use Only. This Device is furnished for information purposes only. Output data — including speed, distance, and position — must not be construed as a guarantee of accuracy.

No Safety-Critical Reliance. The Device must not be used as the sole basis for safety-critical navigation, vehicle operation, stage timing, or any decision where a measurement error could result in personal injury or property damage.

Environmental Limitations. Altura is not liable for errors or incidents arising from GPS signal loss, satellite geometry degradation, multipath interference, or environmental conditions including dense forest canopy, urban structures, tunnels, or atmospheric interference.

3 Assumption of Risk & Limitation of Liability

By purchasing or using the Device or its firmware, the User explicitly and voluntarily assumes all risks associated with motorsport competition and with the use of GNSS-based instrumentation in high-performance environments.

Scope of Disclaimer. To the fullest extent permitted by applicable law, Altura disclaims all liability for personal injury, death, property damage, financial loss, or competitive disadvantage — including lost stage time, event disqualification, or impaired performance — arising from the use or failure of this Device.

Fault Independence. This limitation applies regardless of the legal theory under which a claim is brought, including product malfunction, hardware failure, firmware error, GPS signal degradation, improper installation, or operator misuse.

Installation Responsibility. Altura is not responsible for damages from improper mounting, wiring, or integration with vehicle electrical systems. Do not mount the unit over airbag deployment zones, in a location that obstructs occupant egress, or in any manner that compromises the co-driver's field of vision or access to vehicle controls.

WARNING

Electrical Installation: Installation may require modifications to your vehicle's electrical system. Incorrect installation may cause significant damage to both the Device and the vehicle. Installation by a qualified automotive electrician is strongly recommended. Altura accepts no liability for damage arising from improper electrical installation.

Cap on Liability. In no event shall Altura's total cumulative liability to any User exceed the original purchase price paid for the Device.

4 Firmware & Open-Source License

The Altura Rally Computer runs on open-source firmware distributed under the **GNU General Public License v3.0 (GPL-3.0)**. The full license text is available at <https://www.gnu.org/licenses/gpl-3.0.html> and is included with the firmware source repository.

Open-Source Distribution. Users are encouraged to study, explore, and contribute to the codebase. The complete source code is available through the official Altura firmware repository.

Copyleft Requirement. GPL-3.0 is a strong copyleft license. Any modified version of the firmware must also be distributed under the GPL-3.0 with complete corresponding source code.

Modification Responsibility. Any party modifying or redistributing the firmware must: (a) clearly mark changes from the original; (b) preserve all copyright and GPL-3.0 notices; (c) make complete source code available under GPL-3.0; and (d) ensure full GPL-3.0 compliance. Altura accepts no responsibility for third-party modifications or derivative firmware.

Patent Rights. In accordance with GPL-3.0, contributors grant all recipients a non-exclusive, worldwide, royalty-free patent license under their essential patent claims, to the extent necessary to use the firmware as permitted under GPL-3.0.

"AS-IS" Provision. All firmware is provided "AS-IS" without warranty of any kind. Performance characteristics described in documentation apply only to unmodified firmware on official Altura hardware.

Modified Firmware and Warranty. Hardware warranty coverage described in Section 6 is void if the Device is running firmware modified from the official Altura release. This does not restrict firmware rights under GPL-3.0.

5 Hardware Integrity Policy

Sealed Unit. Opening the unit casing, disassembling the device, or modifying internal hardware immediately and permanently voids all warranty coverage.

Integrity Rationale. The physical reliability of the Device in high-shock and high-vibration conditions depends on factory sealing and assembly tolerances. Tampering with internal hardware undermines the structural integrity for which the unit was designed and tested.

Third-Party Hardware. Altura is not responsible for Device behaviour, data accuracy, or physical damage resulting from hardware modifications, component substitution, or integration with non-approved external hardware.

6 Limited Hardware Warranty

Coverage Period. Altura warrants the hardware against defects in materials and workmanship for **12 months** from the date of original delivery.

Conditions. This warranty applies only to the original Purchaser, is non-transferable, and covers manufacturing defects under normal intended use. It does not cover competition impact, vibration fatigue, improper installation, water ingress, electrical faults from outside the unit, or unauthorized modification.

Sole Remedy. Altura's entire liability under this warranty shall be, at Altura's sole discretion, the repair or replacement of the defective unit. No other remedies, expressed or implied, are available.

Warranty Voidance. The warranty is immediately void upon: (a) opening or disassembling the unit; (b) operating on modified firmware — note that running modified firmware is expressly permitted under GPL-3.0, but voids the hardware warranty only; (c) damage from improper installation or misuse; or (d) any physical modification to the hardware.

7 Return & Refund Policy

Return Window. Returns are accepted within **14 days** of confirmed delivery only. No returns will be accepted after this period.

Eligibility. To be eligible for a refund, the Device must be returned unused, in original condition, and in original packaging. Any unit showing evidence of installation, activation, or physical use is ineligible.

Shipping Costs. The customer is responsible for all return shipping costs and adequate insurance of the shipment. Return shipping will be reimbursed only in the case of a verified manufacturing defect confirmed upon inspection.

Refund Processing. Approved refunds will be issued to the original payment method within 10 business days of confirmed receipt and inspection.

8 Governing Law & Severability

This Agreement shall be governed by and construed in accordance with the laws of the State of Wisconsin, without regard to conflict-of-law principles. If any provision of this Agreement is found unenforceable or invalid, that provision shall be modified to the minimum extent necessary to make it enforceable, and all remaining provisions shall continue in full force and effect.

Acceptance of Terms. Use of the Altura Rally Computer — including powering on the device — constitutes full and binding acceptance of all terms in this Agreement. The firmware is separately governed by the GPL-3.0, the terms of which are independent of and in addition to this Agreement.

9 Third-Party Software Acknowledgments

The Altura firmware incorporates the following third-party open-source components. License notices are preserved in the firmware source repository.

Component	License	Notes
NeoGPS	GPL-3.0	GNSS sentence parsing — primary driver of GPL-3.0 license choice
TinyGPSPlus	LGPL-2.1	GNSS parsing (main branch)
LVGL	MIT	Embedded graphics library
TFT_eSPI	MIT	TFT display driver
FreeRTOS	MIT	Real-time operating system (via ESP32 Arduino core)
Arduino ESP32 Core	LGPL-2.1	Hardware abstraction layer
Adafruit FT6206	BSD/MIT	Capacitive touch (Wokwi simulation branch only)

MIT License Notice (applicable to LVGL, TFT_eSPI, FreeRTOS, Adafruit FT6206):

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Full copyright notices for each component are available in the firmware source repository.

GPS Signal Standards

The Device operates using GPS L1 C/A signals provided by the U.S. Global Positioning System Standard Positioning Service (SPS). The accuracy, availability, and integrity limitations defined by the **U.S. DoD GPS Standard Positioning Service Performance Standard, 5th Edition (April 2020)** apply to all measurements produced by this Device. The U.S. Government does not guarantee GPS signal availability and reserves the right to modify or discontinue SPS signals. The performance limitations in Section 2 are grounded in part on signal-in-space performance bounds established by that standard.

Acknowledgments

The Altura firmware draws on foundational research and published work from the following institutions:

Institution	Contribution
University of Michigan	Optimal estimation theory, Kalman filtering, and sensor fusion methodology
Michigan Technological University	GNSS performance research in forested and high-multipath environments
Free Software Foundation	The GNU General Public License framework enabling open-source distribution
U.S. Dept. of Defense / U.S. Space Force	GPS SPS Performance Standard and continued operation of the GPS constellation

Altura and its authors are grateful to the open-source community whose freely shared code, documentation, and discussion informed this project.

End of Altura Rally Computer License Agreement — v1.0